

Guidelines for the Measurement of Ambient Air Pollutants

Volume-I



Guidelines for Manual Sampling & Analyses



CENTRAL POLLUTION CONTROL BOARD
Ministry of Environment & Forests

Website: <http://www.cpcb.nic.in>

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CENTRAL POLLUTION CONTROL BOARD
Ministry of Environment & Forests
Parivesh Bhawan, East Arjun Nagar, Delhi- 110032
Website: <http://www.cpcb.nic.in>

Guidelines for sampling and analysis for ozone in ambient air (Chemical Method)

1. Purpose

The purpose of this protocol is to provide guidelines for monitoring of ozone in ambient air.

2. Standard

The national ambient air quality standards for ozone is presented in the table

Pollutant	Time Weighted Average	Concentration in Ambient Air	
		Industrial, Residential, Rural and other Areas	Ecologically Sensitive Area (Notified by Central Government)
Ozone (O ₃), µg/m ³	8 Hours *	100	100
	1 Hour **	180	180

* Annual Arithmetic mean of minimum 104 measurements in a year at a particular site taken twice a week 24 hourly at uniform intervals.

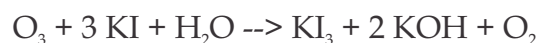
** 24 hourly or 8 hourly or 1 hourly monitored values, as applicable, shall be complied with 98% of the time in a year. 2% of the time, they may exceed the limits but not on two consecutive days of monitoring.

3. Principle of the method

Method 411, Air Sampling and Analysis, 3rd Edition (Determination of oxidizing substances in the atmosphere)

Micro-amounts of ozone and the oxidants liberate iodine when absorbed in a 1% solution of potassium iodine buffered at pH 6.8 + 0.2. The iodine is determined spectrophotometrically by measuring the absorption of tri-iodide ion at 352 nm.

The stoichiometry is approximated by the following reaction:



4. Instrument/Equipment

The following items are necessary to perform the monitoring and analysis of ammonia in ambient air:

- Analytical balance:
- Vacuum pump: Any suction pump capable of drawing the required sample flow rate of 1 to 2 litre per minute
- Calibrated flow measuring device to control the air flow from 1 to 2 l/min.
- Absorber: All glass midget impinger

- Spectrophotometer: Capable of measuring absorbance at 352 nm.
- Glass wares: low actinic glassware must be used for analysis

5. Reagents/Chemicals

All the chemicals should meet specifications of ACS Analytical Reagent grade

- Distilled water
- Absorbing Solution (1% KI in 0.1 M Phosphate Buffer) - Dissolve 13.6 g of potassium dihydrogen phosphate (KH_2PO_4), 14.2 g of disodium hydrogen phosphate (Na_2HPO_4) or 35.8 g of the dodecahydrate salt ($\text{Na}_2\text{HPO}_4 \cdot 12 \text{H}_2\text{O}$), and 10.0 g of potassium iodide in sequence and dilute the mixture to 1 L with water. Keep at room temperature for at least 1 day before use. Measure pH and adjust to 6.8 ± 0.2 with NaOH or KH_2PO_4 . This solution can be stored for several months in a glass stoppered brown bottle at room temperature without deterioration. It should not be exposed to direct sunlight.
- Stock Solution 0.025 M I_2 (0.05N) - Dissolve 16 g of potassium iodide and 3.173 g of re-sublimed iodine successively and dilute the mixture to exactly 500 ml with water. Keep at room temperature at least 1 day before use. Standardize shortly before use, against 0.025 M $\text{Na}_2\text{S}_2\text{O}_3$. The sodium thiosulfate is standardized against primary standard bi-iodate [$\text{KH}(\text{IO}_3)_2$] or potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$).
- M I2 Solution - Pipette exactly 4.00 ml of the 0.025 M Stock solution into a 100 ml low actinic volumetric flask and dilute to the mark with absorbing solution. Protect from strong light. Discard after use.

6. Sampling

Place 10 ml of absorbing solution in a standard impinger and sample for one hour at the flow rate of 1 L/min. Do not expose the absorbing reagent to direct sunlight. After sampling measure the volume of sample and transfer to a sample storage bottle.

7. Analysis

If, appreciable evaporation of the absorbing solution occurs during sampling, add water to bring the liquid volume to 10 ml. Within 30 to 60 minutes after sample collection, read the absorbance in a cuvette at 352 nm against a reference cuvette containing distilled water. Measure the absorbance of the unexposed reagent and subtract the value from the absorbance of the sample.

8. Calibration

8.1. Preparation of Standards

Calibrating Iodine Solution - For calibration purposes exactly 5.11 ml of the 0.001 M I₂ solution (or equivalent volume for other molarity) is diluted with absorbing solution just before use to 100 ml (final volume) to make the final concentration equivalent to 1 µl of O₃/ml. This solution preparation accounts for the stoichiometry described in Section 3 at standard conditions of 101.3 kPa and 25°C. Discard this solution after use.

Obtain a range of calibration points containing from 1 µl to 10 µl of ozone equivalent per 10.0 ml of solution. Prepare by individually adding 1.0, 2.0, 4.0, 6.0, 8.0 and 10.0 mL of the calibrating iodine solution to 10.0 ml volumetric flasks.

Bring each to the calibration mark with absorbing reagent.

Read the absorbance of each of the prepared calibration solutions at 352 nm against distilled water reference

8.2. Standard Curve

Plot a curve absorbance (Y axis) versus concentration (X axis). Draw a line of best fit and determine the slope. The reciprocal of slope gives the calibration factor (CF).

9. Calculation

$$C(\text{O}_3 \mu\text{g}/\text{m}^3) = (A_s - A_b) \times \text{CF} \times 1.962 / V_a$$

Where,

C O₃ = Concentration of Ozone in µg/m³

A_s = Absorbance of sample

A_b = Absorbance of reagent blank

CF = Calibration factor

V_a = Volume of air sampled in m³

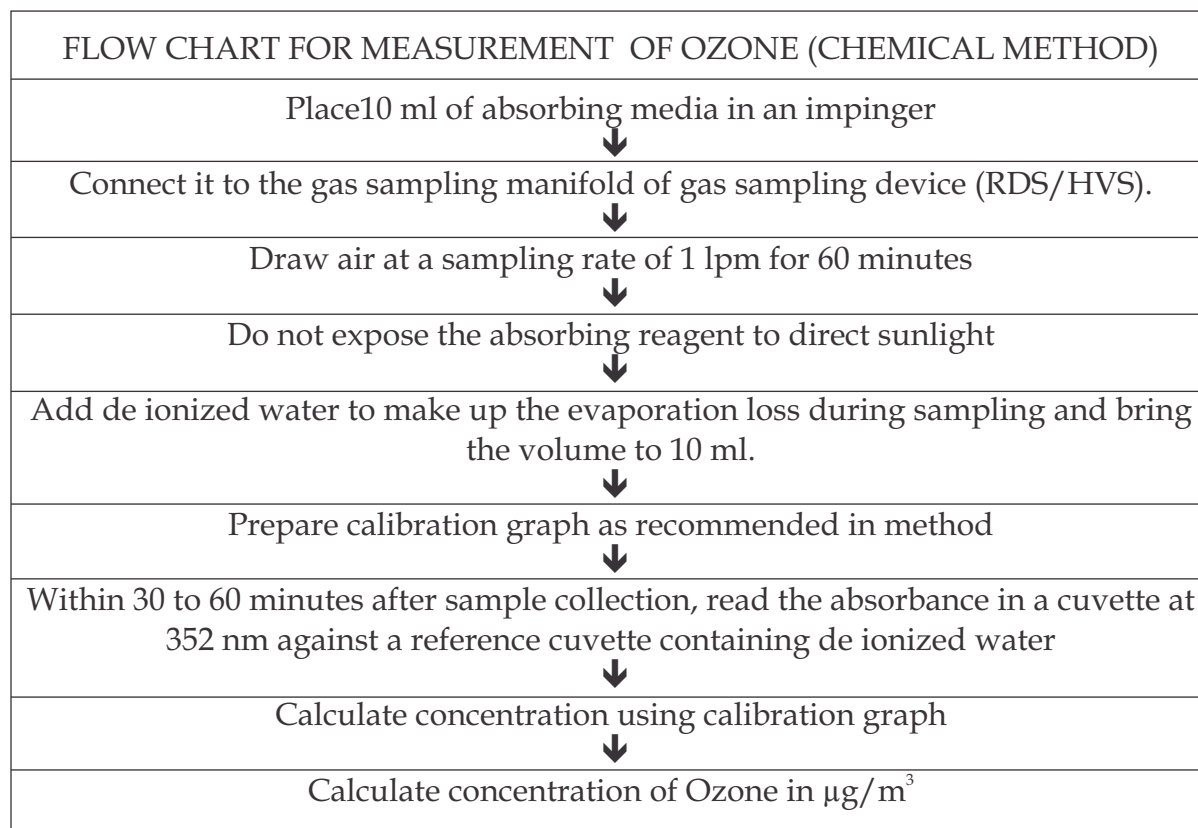
1.962 = Conversion factor, µl to µg

10. Quality Control

Quality Control (QC) is the techniques that are used to fulfil requirements for quality. The QC procedures for the air sampling and monitoring sections of this protocol include preventative maintenance of equipment, calibration of equipment, analysis of field blanks and lab blanks.

11. Reference

Method 411, Air Sampling and Analysis, 3rd Edition (Determination of oxidizing substances in the atmosphere)



DATA SHEET FOR OZONE IN AMBIENT AIR

Monitoring Duration	Average Flow Rate (lpm)	Total Sampling Time (Minutes)	Total Volume of Air Sampled (lit.)	Volume of Sample (ml)	Absorbance (Blank)	Absorbance (Sample)	Conc. ($\mu\text{g}/\text{m}^3$)
0600-0700							
0700-0800							
0800-0900							
0900-1000							
1000-1100							
1100-1200							
1200-1300							
1300-1400							
1400-1500							
1500-1600							
1600-1700							
1700-1800							
1800-1900							
1900-2000							
2000-2100							
2100-2200							
2200-2300							
2300-2400							
2400-0100							
0100-0200							
0200-0300							
0300-0400							
0400-0500							
0500-0600							

CENTRAL POLLUTION CONTROL BOARD

Data Sheet for Ozone

Month:

Station:

Duration Date ↓	06- 07	07- 08	08- 09	09- 10	10- 11	11- 12	12- 13	13- 14	14- 15	15- 16	16- 17	17- 18	18- 19	19- 20	20- 21	21- 22	22- 23	23- 24	24- 01	01- 02	02- 03	03- 04	04- 05	05- 06

Concentrations in $\mu\text{g}/\text{m}^3$

**CENTRAL POLLUTION CONTROL BOARD
AIR LAB DIVISION**

FIELD DATA SHEET (BaP/PAH MONITORING)

Monitoring Location Details:

Sheet No.-----

Date & time of start	Date & time of close	Run Time in Minutes
Instrument Make & S.No	Average Flow Rate	Total Air Volume
Filter Paper No	Final Weight (gm)	Initial Weight (gm)
Net Weight (gm)	PM10 Conc. (ug/m ³)	Special Weather Note (Rainy/cloudy/Sunny/ Windy/stormy Day)
PI fill which ever Shift applicable		

Monitoring team members (Names with Signatures & Dates):

Time	Mano- meter Reading m ³ /min	Timer Reading	Time	Mano- meter Reading m ³ /min	Timer Reading	Time	Mano- meter Reading m ³ /min	Timer Reading
6.00			14.00			22.00		
7.00			15.00			23.00		
8.00			16.00			24.00		
9.00			17.00			01.00		
10.00			18.00			02.00		
11.00			19.00			03.00		
12.00			20.00			04.00		
13.00			21.00			05.00		
14.00			22.00			06.00		
Avg. Flow m3/min			Avg. Flow m3/min			Avg. Flow m3/min		

1. _____ 2. _____ 3. _____

Project Coordinator (Name with Signature): _____

CENTRAL POLLUTION CONTROL BOARD
AIR LAB DIVISION
BaP/PAH ANALYSIS REPORT

Report S. No. _____

Date: _____

S.No.	Sampling Location	Sample details	Date of Sampling	Time of sampling	Benzo(a)Pyrene (ng/m ³)	TPAH (ng/m ³) (if analysed)
1.						
2.						
3.						
4.						
5.						
6.						

Note: Sample analyzed using Capillary GC-FID.

(Analyst)

(Supervisor)

(I/C –Laboratory)

CENTRAL POLLUTION CONTROL BOARD

DATA SHEET FOR PARTICULATE MATTER (Size less than 10 μm) or PM_{10}

Station:		Date:	
Shift			
Monitoring Duration			
Filter Paper No.			
Hourly Flow Rate (m^3/minute)			
Average Flow Rate (m^3/minute)			
Total Operation Time (Minutes)			
Initial Weight of Filter Paper (gms.)			
Final Weight of Filter Paper (gms.)			
Dust Contents (gms.)			
Total Volume of Air Sampled (m^3)			
Concentration ($\mu\text{g}/\text{m}^3$)			
24 Hourly Average SPM ($\mu\text{g}/\text{m}^3$):			
Remarks:			
Name & Signature of Official on Duty:			
Analysed by:			

CENTRAL POLLUTION CONTROL BOARD

Parivesh Bhawan, East Arjun Nagar, Delhi -110 032

Trace Metals Analysis Report

1. Report no. & issue date :
2. Name of the Project :
3. Sample matrix :
4. Date & time of sample collection :
5. Samples collected by :
6. Date & time of sample receipt :
7. Date of sample analysis :
8. Sample registration no. & date :
9. Sampling plan reference :
10. Test method reference :
11. Report sent to (Name & Address) :

S. No.	Sample Code	Pb ($\mu\text{g/ml}$)	Ni ($\mu\text{g/ml}$)	As ($\mu\text{g/ml}$)

Statement:

1. The results relate only to the samples tested.
2. The report shall not be reproduced except in full, without the written approval of the laboratory.

Analyst

Supervisor

In-Charge

CENTRAL POLLUTION CONTROL BOARD
Parivesh Bhawan, East Arjun Nagar, Delhi -110 032

Analysis report of Metals

Month:

Station:

Date	24 hour average concentration		
	Lead ($\mu\text{g}/\text{m}^3$)	Nickel (ng/m^3)	Arsenic (ng/m^3)

Analyst

Supervisor

Incharge

Ambient Air Quality Data Reporting Format (Annual)

Monitoring Agency :

Year :

Name of City :

Name of Station :

Station Code :

Area :

Parameters	Unit	No. of 24 Hourly observations	Annual Mean	Measurement Method #	
Sulphur dioxide (SO ₂)	µg/m ³			Improved West and Gaeke Method	
				Ultraviolet Fluorescence	
				Any Other	
Nitrogen dioxide (NO ₂)	µg/m ³			Jacob & Hochheiser modified (NaOH-NaAsO ₂) Method	
				Gas Phase Chemiluminescence	
				Any Other	
Ammonia (NH ₃)	µg/m ³			Chemiluminescence	
				Indophenol method	
				Any Other	
Particulate Matter, PM ₁₀	µg/m ³			Gravimetric	
				Beta attenuation	
				TEOM	
				Any Other	
Particulate Matter, PM _{2.5}	µg/m ³			Gravimetric	
				Beta attenuation	
				TEOM	
				Any Other	
Lead (Pb)	µg/m ³			AAS/ICP Method after sampling on EPM 2000 or equivalent filter paper	
				ED-XRF using Teflon filter	
				Any Other	
Nickel (Ni)	ng/m ³			AAS/ICP Method after sampling on EPM 2000 or equivalent filter paper	
				Any Other	
Arsenic (As)	ng/m ³			AAS/ICP Method after sampling on EPM 2000 or equivalent filter paper	
				Any Other	
Benzo (a) Pyrene (BaP)	ng/m ³			Solvent extraction followed by HPLC/GC analysis	
				Any Other	
Benzene (C ₆ H ₆)	µg/m ³			Gas Chromatography (GC) based continuous analyzer	
				Adsorption and desorption followed by GC analysis	
				Any Other	

Tick (✓) mark the method followed

(Checked & Compiled by)

(Authorized Signatory)

Area :

Note: 24 hourly Data of Benzene required for calculation of Annual average

(Authorized Signatory)

Station Code :

[illegible]

Tick (✓) mark the method followed

(Checked & Compiled by)

(Authorized Signatory)

Ambient Air Quality Data Reporting Format (24 Hours)

Monitoring Agency :

Month & Year :

Name of City :

Name of Station :

Station Code :

Area : :

[illegible]

Note: 24 hourly Data of Nickel, Arsenic & BaP required for calculation of Annual average

(Checked & Compiled by)

(Authorized Signatory)



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 Fax 0364-2520805

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A Clean PARIVESH for all is our goal